

Quantum Treatment of Power Law based Signaling in Kuramoto Oscillator Networks

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Abstract—We propose a framework linking quantum entanglement, power-law signal dynamics, and neural-network learning through the lens of Kuramoto-style oscillator networks. In this approach, phase entanglement is interpreted as metaheuristic coordination within quantum systems, effectively modeled by Boltzmann machines with power-law governed signal structures. Reduced Boltzmann Machines (RBMs) simulate polymer ring condensates and entanglement networks, capturing multipartite correlations in an extended configuration space. To explore these ideas experimentally, we investigate phase-oscillator LED networks, where time-series analysis and Takens embedding reveal that local clusters stabilize into coherent attractors while interacting across the network. Notably, a “bridging” phenomenon emerges, with certain clusters acting as pivots between others, effectively encoding memory-like roles. These dynamics mirror metastable basins in RBM-inspired energy landscapes, suggesting that the system forms a scale-invariant, path-integral-like architecture capable of mapping inputs to outputs through phase-oscillator interactions. Our results highlight a configuration-space-driven mechanism for collective coherence, bridging classical synchronization phenomena and quantum-inspired network architectures, with implications for quantum computation, networked learning, and emergent behavior in complex systems.

I. INTRODUCTION

Entanglement is often considered the cornerstone of quantum mechanics, distinguishing it fundamentally from classical physics [3]. It enables nonlocal correlations and underpins many of the proposed advantages of quantum computation and communication [4]. However, modeling entangled systems—particularly in dynamically interacting, many-body settings—remains a central challenge in quantum theory.

The Kuramoto model [5], initially developed to describe synchronization in classical oscillator systems, has found unexpected utility in describing collective behavior in complex systems, including biological and quantum contexts [6]. Metaheuristics, which are optimization frameworks inspired by biological, physical, or social systems, are increasingly being used to approximate high-dimensional quantum systems where exact methods fail.

In this work, we investigate the idea that entanglement, particularly when represented in terms of von Neumann entropy for bipartite systems, may be thought of not only as a resource but as a *metaheuristic signaling mechanism*. In this picture, power-law behaviors — which dominate many physical systems including the inverse-square law of light propagation — are the natural carriers of such signals. This allows us to bridge from classical communication dynamics to quantum computational representations via Boltzmann machines, especially their reduced forms that mirror the modularity of polymer rings or entanglement clusters.

We build upon previous work where Boltzmann machines were used as thermodynamically inspired learners [7], and suggest that their configuration spaces can be naturally extended to represent entanglement condensates — correlated clusters within quantum systems that obey nonlocal probabilistic constraints and evolve under power-law driven dynamics.

II. SIGNALLING AS AN ABSTRACT INTERACTION PRIMITIVE

Rather than deriving signaling behaviour from explicit optical or physical attenuation laws, we adopt an abstract view in which signaling strength is treated as a generic interaction primitive between nodes in a networked system. In this formulation, the precise physical carrier of the signal—whether optical, acoustic, chemical, or otherwise—is not specified. Instead, signaling is characterized solely by its functional role in mediating influence, coordination, and synchronization across spatially separated elements.

This abstraction allows the signaling mechanism to be interpreted independently of any particular physical realization, while remaining compatible with a wide class of decay profiles and interaction kernels. The essential requirement is that interaction strength diminishes with increasing separation in the system’s configuration space, thereby enforcing locality and enabling structured collective behaviour.

By framing signaling in this manner, we avoid committing to a specific physical propagation model and instead focus on how such interactions give rise to adaptive, self-organizing, and optimization-driven dynamics. This perspective is particularly well suited for metaheuristic analysis, where the emphasis lies on emergent behaviour rather than microscopic physical detail.

The subsequent sections build upon this abstract signaling concept to examine how exploration–exploitation balance, synchronization, and least-action principles naturally arise within both classical and quantum-inspired frameworks.

III. POWER-LAW BASED QUANTUM NETWORK SIGNALING

The most basic signaling we can construct that works for quantum as well as classical light is one based on some sort of power law for signalling. Light signalling can of course be represented by a power law. Light signal transmission, like light in general, obeys the inverse-square law in its propagation through space. That is to say, the intensity of the light, I , is inversely proportional to the square of the distance from the light source.

$$I(r) = \frac{I_0}{r^2}$$

Therefore the light signal gets weaker and weaker as the distance increases.

The meta-heuristic algorithm therefore must contain a function which monotonically decreases its signalling power between each transceiver node with distance, r , under a discrete light absorption coefficient for the physical medium Y . [3]

Under a given physical medium, the Gaussian form of the light intensity is then determined by:

$$I(r) = I_0 e^{-\gamma r^2}$$

We can use this equation of course to find the light absorption coefficient itself:

$$\begin{aligned} \frac{I(r)}{I_0} &= e^{-\gamma r^2} \Rightarrow \ln \left(\frac{I(r)}{I_0} \right) = -\gamma r^2 \Rightarrow \\ \gamma &= \frac{-\ln \left(\frac{I(r)}{I_0} \right)}{r^2} \end{aligned}$$

The generic power law signalling function of is then written to be monotonically decreasing:

$$\beta = \beta_0 e^{-\gamma r^m}$$

Where r is the distance between the different nodes. m is the power law parameter for the signal, for light signals it is $m = 2$ (for signals that use audio, chemical, etc, power laws it will be different) Where β_0 is the emitted signal at $r = 0$

The distance between any two nodes, i and j at $\mathbf{X}_i$ and $\mathbf{X}_j$ respectively, is the Cartesian distance as follows:

$$r_{ij} = \|\mathbf{X}_i - \mathbf{X}_j\| = \sqrt{\sum_{k=1}^d (x_{i,k} - x_{j,k})^2}$$

Where $x_{i,k}$ is the (k) th component of the spatial coordinate $\mathbf{X}_i$ of the (i) th node.

Where d is the number of dimensions.

The time evolution of each individual node, i , over its signalling cycle is governed by the fact each typical node, i , is coupled to the brightest node it sees, j , by the following equation:

$$X_j = X_i + \left(\beta_0 e^{-\gamma r^2} \right) (X_i - X_j) + \alpha$$

The second term is the signalling term, for most cases in implementation $\beta_0 = 1$

The third term is the randomization term for a possible discrete coupling path which uses a randomization parameter, α , which for most cases in implementation is distributed in the domain of $[0, 1]$.

An interesting way to write this equation is:

$$X_j \left(1 + \beta_0 e^{-\gamma r^2} \right) = \left(1 + \beta_0 e^{-\gamma r^2} \right) (X_i) + \alpha$$

Which when tidied up further reads:

$$(X_i - X_j) = \frac{\alpha}{(1 + \beta_0 e^{-\gamma r^2})}$$

We can then define the signalling Activation Function, A :

$$A(X_{i,j}) = \left(1 + \beta_0 e^{-\gamma r^2} \right)^{-1}$$

Which reduces our equation to a more fundamental signal field representation:

$$\Phi(X_{i,j}) = A(X_{i,j}) \cdot \alpha$$

Based on these formulae, we have the corresponding rules:

- Each node in the field will emit and absorb discrete light signals equally.
- The signals are proportional to the intensity of the light emitted, which both decrease in proportion to increasing distance between the nodes.
- For any 2 flashing nodes in the field, the signal absorbed with the highest intensity will induce the strongest coupling
- If there is no signal detected with higher intensity than any one particular node, it will designate itself as being isolated and signal randomly
- The light intensity of any signaling node is determined by the landscape of the field, itself determined by the nature of the activation function itself.

IV. WHY IS THIS TYPE OF SIGNALING META-HEURISTIC? - DEFINING THE CRITERIA OF THE META-HEURISTIC BEHAVIOR:

Heuristic algorithms typically intend to find a good solution to an optimization problem by 'trial-and-error' in a reasonable amount of computing time. Here 'heuristic' means to 'find' or 'search' by trials and tallying the hits and misses. There is no guarantee to find the best or optimal solution, though it might be a better or improved solution than an educated guess.

Any reasonably good solution, often suboptimal or near optimal, would be good enough for such problems. Broadly speaking, local search methods are heuristic methods because their parameter search is focused on the local variations, and the optimal or best solution can be well outside this local region. However, a high-quality feasible solution in the local region of interest is usually accepted as a good solution in many optimization problems in practice if time is the major constraint.

V. ABSTRACTION - THE PATH INTEGRAL INTERPRETATION OF QUANTUM MECHANICS:

Path-integral theory is a rigorous formulation of quantum theory that uses a particles path across space-time instead of wave functions. It has particular use in quantum many-body systems as explored in quantum field theory.

Imagining a particle at some initial position in the (x,y)-plane, we want to know what path it will take to some final position. By the classical least action principle, the particle will take a path between the two positions that costs the least energy. But, if the particle is a quantum particle, it's not really localized at a point. Instead, the particle is a wave... and it doesn't take one path from the initial position to the final position, it takes all possible paths.

The basic idea then is to construct matrix elements of the time-evolution operator by summing all possible paths between two points and weighting the paths by the classical principle of least action. Remarkably, this gives identical results to solving the Schrödinger wave equation, but is much more concise a model.

The Hamiltonian generates the time evolution of quantum states. If $|\psi(t)\rangle$ is the state of the system at time t , then

$$H|\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle$$

This equation is the Schrödinger wave equation

From the Schrödinger wave equation we of course get the time evolution operator to work with.

Given the state at some initial time ($t = 0$), we can solve it to obtain the state at any subsequent time. In particular, if H is independent of time, then

$$|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle.$$

By the homomorphic property of the functional calculus, the operator

$$U = e^{-iHt/\hbar}$$

This is the time evolution operator, or propagator, of a closed quantum system.

Each path the particle takes is equivalent to a probability wave as described by the Schrödinger wave equation. Waves are in fact represented here in the complex plane as being circles traced out by the wave-form.

This phenomenon is inherently related to the nature of the complex plane, where the complex number $z = \exp(i\theta) = \cos(\theta) + i\sin(\theta)$ acts like a unit vector with a phase angle θ in the complex plane

Euler's Equation can then be used to establish the fundamental relationship between the trigonometric functions and the complex exponential function. Euler's formula states that, for any real number x :

$$e^{ix} = \cos x + i\sin x$$

Where e is the base of the natural logarithm, i is the imaginary unit, and $\cos$ and $\sin$ are the trigonometric functions cosine and sine respectively, with the argument x given in radians.

The path of least action arises since the unit vector, z , tend to cancel out when the phase angles are out of phase

with each other, and the vectors tend to add together when the phase angles are in phase with each other:

By taking each unit vector along probability wave represented by the path the particle takes, we form a series of arrows and make them oscillate around the complex unit circle at a frequency based on how difficult it is for a particle to travel through a given point-i.e. the harder the path to travel, the slower the arrow spins. (oscillating with frequency equal to the action.)

As discussed before that the particle takes all paths between the initial state and the final state. So we make our arrows follow each path and rotate them as we go along, as shown below.

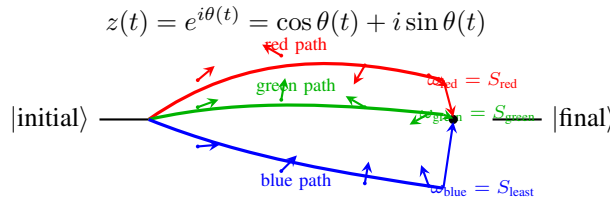


Fig. 1. Feynman sum-over-histories representation. Each possible path (red, green, blue) carries a unit complex vector $z(t) = e^{i\theta(t)}$ rotating with a frequency proportional to its action S . Paths with larger action rotate more rapidly and cancel by destructive interference, while the least-action path rotates slowest and contributes most to the final amplitude.

Hence, in the path integral interpretation, the clockwork of unit vector arrows are along every possible path along the initial and final state (in this case 3 possible paths, red, green and blue). The paths are different only insofar as the arrows are rotated at different frequencies equal to the deviation from the path of least action, S . In other words the arrows are rotated with a frequency equal to that of the action imparted to deviate the particle from that of the least action possible. Therefore the path of least action itself should have the lowest frequency of unit vector rotation, where an increase in frequency corresponds to an increase in action.

VI. VIEWING META-HEURISTIC SYNCHRONIZATION USING THE PATH INTEGRAL:

The application of the path integral formalism in quantum mechanics can be extended to describe synchronization phenomena in entangled networks, cavities, and reservoirs. We discussed this in our work on using Lagrangian mechanics for systems undergoing synchronization and its connection to least-action principles [12], [13].

The resemblance of the unitary time-evolution operator to the monotonically decreasing signaling function is obvious in their respective mathematical forms [13].

By replacing the concept of least action with a least signaling term, we investigate how the path integral representation can be applied to study synchronization dynamics in entangled networks, drawing parallels with coupled oscillators in phase space models such as the Kuramoto model, as well as its relation to algorithms, such as the firefly algorithm for potential use in optimization in networks.

For one thing, the metaheuristic signaling algorithms is readily compatible with Lagrangian mechanics which are inherently used for discrete particles each with a finite number of degrees of freedom. Our Signaling Lagrangian, L_f , for each signaling node are in-effect integrated across the time-domain in the path it takes to achieve synchronization, hence leading to the total Signaling Action S_f , represented here as being the phase of each path being determined by $\int L_f dr$, for that trajectory, where L_f is the Lagrangian.

$$S_f = \int L_f dr = \ln \left(\frac{I(r)}{I_0} \right) = -\gamma r^2$$

The frequency of signaling is then, as an oscillation in the complex plane:

$$e^{S_f}$$

However, we can investigate go further and compare the probabilistic representation of the path integral theory with the description of the algorithm as a system of discontinuous pas-coupling

Using this in the path-integral view we can represent the signaling procedure as:

$$\langle \bigcirc | \beta_0 e^{-\gamma r^2} | \bigcirc \rangle = \text{Amplitude for Signal}$$

In this convention, the signaling action, S_f , which is a real operator is defined as being essentially characteristic of the physical and environmental parameters of the system - namely the light absorption coefficient and the power law over the distance by which it is subject to

Since each deviation from the path of least action is, just like in the case with a quantum particle, proportional to the action imparted on the system, the signal activation function will activate with a frequency proportional to the action.

Therefore, the path of least possible action, which occurs during complete internal synchronization, should have the lowest frequency of signal function activation.

An increase in frequency of signal activation indicates a deviation from the path of least possible signaling action.

Therefore, when using any metaheuristic algorithm, when looking for any deviations, i.e. faults, between the coupled oscillators when internal synchronization is achieved we should just have to look for any changes in the frequency of the signaling between the different nodes.

This also explains how the self-organising system restores itself - as a fault is equivalent to a deviation from the principle of least action, which is energetically unfavourable.

VII. THE BOLTZMANN MACHINE AS A KURAMOTO NETWORK CONFIGURATION

In classical statistical mechanics, the configuration space of a system is the set of all accessible microstates. In quantum systems, especially those exhibiting multipartite entanglement, this space must be extended to include nonlocal correlations encoded in the full Hilbert-space structure. We propose to model this extended configuration space using Boltzmann machines—specifically Reduced Boltzmann Machines (RBMs)—where network couplings encode the structure of quantum correlations.

A. RBMs as Encoders of Entangled Clusters

In this framework:

- visible units represent classical measurements or subsystem observables,
- hidden units represent virtual qubits or entangled clusters,
- network weights encode nonlocal quantum correlations.

The RBM energy function

$$E(\mathbf{v}, \mathbf{h}) = - \sum_i a_i v_i - \sum_j b_j h_j - \sum_{ij} W_{ij} v_i h_j, \quad (1)$$

defines an emergent “quantum-inspired” energy landscape. Stochastic activation mimics quantum measurement statistics, while the temperature parameter encodes effective decoherence.

Within this picture, *entanglement condensates*—coherent regions of Hilbert space—correspond to attractors or metastable basins in the RBM landscape.

B. Ring Topology and Quantum Field Theory Analogies

Arranging RBM modules in a cyclic ring topology parallels cyclic boundary conditions in path-integral formulations of quantum mechanics [8]. Such structures also reflect the braid-group representations used in topological

quantum computing [9]. Transitions between metastable RBM states may be interpreted as topologically protected tunneling events, forming a quantum-inspired memory space.

C. Kuramoto Interpretation

A classical Kuramoto network satisfies

$$\dot{\theta}_i = \omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i), \quad (2)$$

with collective synchronization measured by the order parameter

$$Re^{i\Psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}. \quad (3)$$

In our analogy:

- visible units act as oscillator sites,
- hidden units represent higher-order couplings,
- weights W_{ij} correspond to effective phase-coupling strengths.

A quantum generalization replaces phases with operator expectation values:

$$\frac{d}{dt} \langle \hat{\theta}_i \rangle = \omega_i + \sum_j K_{ij} \text{Im} \left[e^{i(\langle \hat{\theta}_j \rangle - \langle \hat{\theta}_i \rangle)} \rho \right]. \quad (4)$$

This motivates the use of optical Kuramoto networks as classical analogs of RBM-encoded entanglement structures.

VIII. TAKENS’ EMBEDDING AND NONLINEAR TIME SERIES ANALYSIS OF LED OPTICAL SYNCHRONIZATION

A. A Note on Takens’ Embedding

Takens’ theorem states that a smooth dynamical system with attractor dimension d can be reconstructed from a scalar observable $x(t)$ using delay vectors:

$$\mathbf{x}(t) = [x(t), x(t + \tau), \dots, x(t + (m - 1)\tau)], \quad (5)$$

provided that the embedding dimension satisfies $m > 2d$. In this study we use:

$$\tau = 5, \quad m = 3,$$

which allows for visualization in 3D.

In the quantum context, delay embeddings may be applied to classical measurement records such as homodyne currents or photon counts, enabling partial reconstruction of quantum network dynamics.

B. Experiment: Kuramoto LED Network

A spatially distributed network of red and blue LEDs was recorded while undergoing Kuramoto-style coupling. Frame-by-frame OpenCV processing extracted normalized intensities for each color, producing two scalar time series.

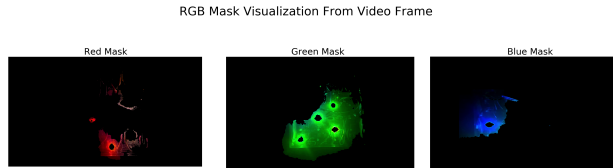


Fig. 2. Snapshot of the LED network undergoing Kuramoto-type optical synchronization.

C. Time Series Overview

The raw intensity time series for the red and blue channels is shown in Fig. 3.

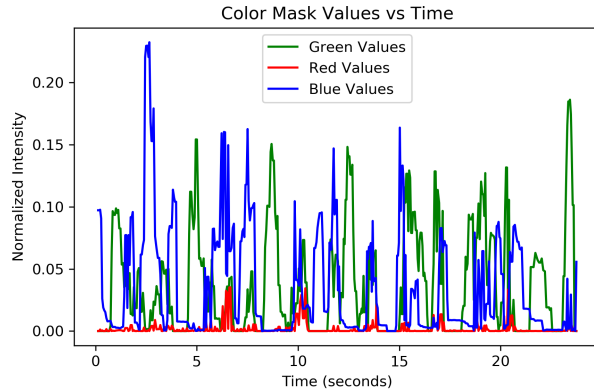


Fig. 3. Intensity time series for red and blue LED channels, showing synchronization onset and regime transitions.

D. Reconstructed Phase Space

Applying Takens' embedding to the red and blue time series yields 3D attractor reconstructions:

These visualize the dynamical structure of the synchronized optical network, similar to attractors seen in chaotic laser-based communication systems.

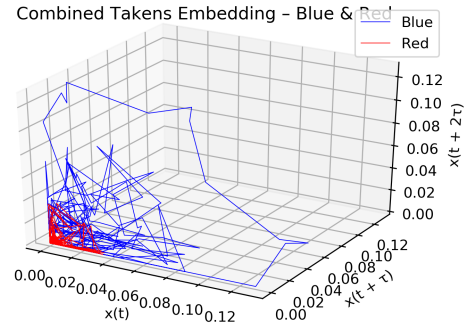


Fig. 4. Combined red-blue embedding revealing joint attractor structure and cross-channel phase coupling.

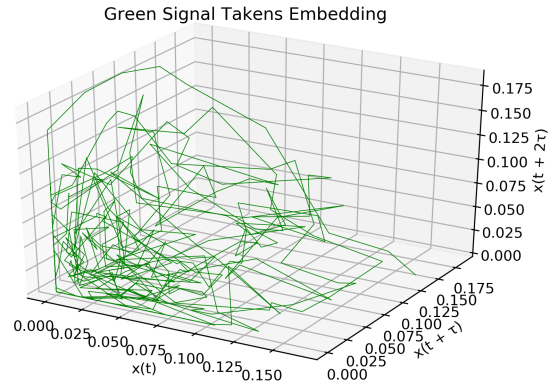


Fig. 5. Derived configuration-space embedding (green channel) capturing the collective behavior of the optical network.

E. Interpretation and Connection to Quantum/Kuramoto Networks

Takens reconstruction exposes the geometry of the system's synchronization manifold. In quantum synchronization scenarios, similar embeddings may be applied to measurement records encoding reduced density operator evolution. Thus, the LED Kuramoto network serves as a classical analog of:

- RBM-encoded entanglement dynamics,
- phase-coupled quantum networks,
- chaotic synchronization structures used in secure optical communications.

This supports the broader claim that Boltzmann-machine networks and Kuramoto-type oscillators offer a unified descriptive framework for both classical chaotic synchronization and quantum-network coherence.

IX. LAGGING SYNCHRONIZATION IN MEDIATED OSCILLATOR NETWORKS

A. Three-node Kuramoto Chain: Source–Channel–Client

Consider a directed synchronization chain comprising three phase oscillators: a sender node (**Alice**), an intermediary memory/channel node (**Charlie**), and a receiver node (**Bob**). Each node is characterized by a phase variable $\phi_i(t)$ and natural frequency ω_i . The system obeys a directed Kuramoto-type model:

$$\frac{d\phi_A}{dt} = \omega_A, \quad (6)$$

$$\frac{d\phi_C}{dt} = \omega_C + K_{AC} \sin(\phi_A - \phi_C), \quad (7)$$

$$\frac{d\phi_B}{dt} = \omega_B + K_{CB} \sin(\phi_C - \phi_B). \quad (8)$$

Here:

- ϕ_A , ϕ_C , and ϕ_B are the phases of Alice, Charlie, and Bob respectively.
- K_{AC} and K_{CB} are coupling strengths from Alice to Charlie and Charlie to Bob.

Unlike the globally coupled Kuramoto model, this topology is strictly *causal and directional*: Alice influences Charlie, and Charlie influences Bob, but there is no back-action.

B. Phase Lag and Memory Effects

In this configuration, synchronization between Alice and Bob cannot occur instantaneously. Charlie acts as both a *memory* and a *phase buffer*, inducing a delay. This results in a temporal lag:

$$\phi_A(t) > \phi_C(t) > \phi_B(t), \quad (9)$$

during the transient dynamics as synchronization unfolds. The steady-state phase differences $\Delta_{AC} = \phi_A - \phi_C$ and $\Delta_{CB} = \phi_C - \phi_B$ may stabilize depending on the coupling strengths and frequency detuning.

This lagging chain represents a coarse-grained model of phase information propagation across a non-instantaneous communication channel. It highlights the importance of intermediaries in both physical and informational synchrony.

C. Super-oscillator Networks: Alice–Charlie–Bob

We may generalize the three-node model by treating Alice and Bob as *networks* of many oscillators, renormalized into collective (average) phases:

$$\Phi_A(t) = \frac{1}{N_A} \sum_{i \in A} \phi_i(t), \quad \Phi_B(t) = \frac{1}{N_B} \sum_{j \in B} \phi_j(t)$$

Charlie likewise comprises a mesoscopic network of N_C oscillators whose internal dynamics can reflect complex memory effects such as echo, dispersion, or decoherence.

Thus, the effective synchronization between Alice and Bob becomes mediated through the emergent dynamics of Charlie's collective phase $\Phi_C(t)$, potentially exhibiting:

- **Lagging synchronization:** due to finite propagation or processing time.
- **Chimera effects:** where coherence and incoherence coexist in Charlie.
- **Non-trivial delays:** reflecting internal structure and resonance modes in Charlie.

D. Schematic Representation

A conceptual schematic of this setup is shown below:

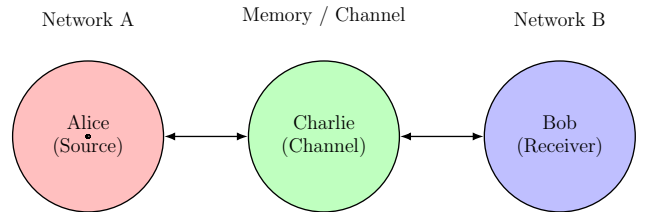


Figure: Alice and Bob each represent synchronized networks modeled as super-oscillators. Charlie is a mediating network acting as a channel with lagging phase dynamics. Arrows represent unidirectional phase influence.

This model is structurally analogous to quantum communication systems, where two entangled nodes are connected via an intermediate quantum channel. However, unlike mechanical couplings in classical systems (e.g. Huygens' pendulums), the coupling in quantum networks is not spatial but exists in a more abstract configuration space — a phase space — echoing the attractor-based synchronization observed in chaotic systems.

X. KURAMOTO CHAIN NETWORK SIMULATION

This section describes the design and implementation of a Kuramoto oscillator chain simulation consisting of three interconnected subsystems: **Alice**, **Charlie**, and **Bob**. The simulation evaluates how oscillator phases evolve and synchronize under varying coupling strengths and structural configurations.

A. Network Architecture

The network comprises:

- **Alice:** A 3×3 2D grid network.
- **Charlie:** A small-world network generated using the Watts-Strogatz model.
- **Bob:** A 3×3 2D grid network.

Each node i is assigned an intrinsic frequency ω_i and a phase θ_i . Coupling between nodes can be symmetric or asymmetric depending on simulation parameters.

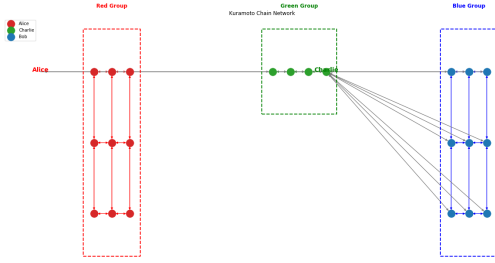


Fig. 6. Kuramoto chain topology showing intra-group (colored edges) and inter-group (gray directional edges) connections between Alice, Charlie, and Bob.

B. Phase Dynamics Formulation

The discrete-time dynamics of each oscillator in the network are governed by a delayed Kuramoto model. We first present the scalar form for each node, followed by a compact matrix formulation for the entire system.

1) Scalar Form:

$$\theta_i(t+\Delta t) = \theta_i(t) + \Delta t \left[\omega_i + \frac{1}{k_i} \sum_{j \in \mathcal{N}(i)} K_{ij} \Im \left(e^{i(\theta_j(t-\tau_{ij}) - \theta_i(t))} \right) \right] \quad (10)$$

Where:

- $\theta_i(t)$ is the phase of node i at time t
- ω_i is the natural frequency of node i
- K_{ij} is the coupling strength from node j to node i
- k_i is the degree (number of neighbors) of node i

- τ_{ij} is the delay from node j to i (directional: Charlie $\rightarrow$ Bob only)
- Δt is the time integration step size
- $\Im(\cdot)$ denotes the imaginary part of a complex number

2) *Matrix Form:* The dynamics for all N nodes can be written compactly as:

$$\boldsymbol{\theta}(t+\Delta t) = \boldsymbol{\theta}(t) + \Delta t \left[\boldsymbol{\omega} + D^{-1} \cdot \Im \left(K \circ \exp \left(i(\boldsymbol{\theta}(t-\boldsymbol{\tau}) - \boldsymbol{\theta}(t)) \right) \mathbf{1} \right) \right] \quad (11)$$

Where:

- $\boldsymbol{\theta}(t) \in \mathbb{R}^N$ is the vector of all node phases at time t
- $\boldsymbol{\omega} \in \mathbb{R}^N$ is the vector of natural frequencies
- $K \in \mathbb{R}^{N \times N}$ is the matrix of pairwise coupling strengths
- $\boldsymbol{\tau} \in \mathbb{R}^{N \times N}$ is the matrix of pairwise delays
- $D = \text{diag}(k_1, \dots, k_N)$ is the diagonal degree matrix
- $\circ$ denotes the Hadamard (element-wise) product
- $\exp(i(\cdot))$ is applied element-wise to vectors
- $\mathbf{1}$ is a vector of ones used for summing across neighbors
- $\Im(\cdot)$ extracts the imaginary part element-wise

C. Simulation Workflow

1) Initialization:

- Create three subgraphs: grid (Alice, Bob) and small-world (Charlie)
- Assign random initial phases $\theta_i \in [0, 2\pi]$ and $\omega_i \sim \mathcal{U}(0.95, 1.05)$
- Create inter-group directional edges with tunable coupling weights

2) Phase Update:

- Compute aggregated group phases: $\bar{\theta}_{\text{Alice}}, \bar{\theta}_{\text{Charlie}}, \bar{\theta}_{\text{Bob}}$
- Use stored phase history for delay modeling between Charlie and Bob
- Apply discrete Kuramoto dynamics with normalized coupling inputs

3) Visualization:

- Network topology and node coloring based on $\sin(\theta_i)$
- Evolution of group phase averages on a unit circle
- Time-series plots of:
 - Group average phases

- Global order parameter

$$r(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right|$$

- Adjacency matrix heatmap sorted by Fiedler vector for spectral graph insights

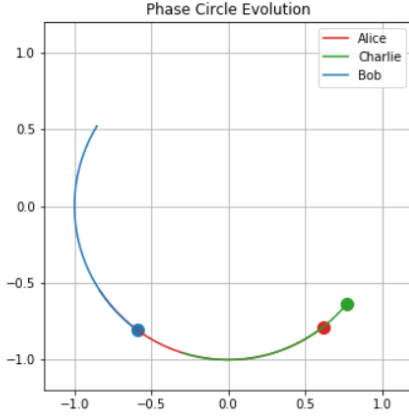


Fig. 7. Phase circle trajectories of Alice, Charlie, and Bob over time. The latest phase state is highlighted with a large dot.

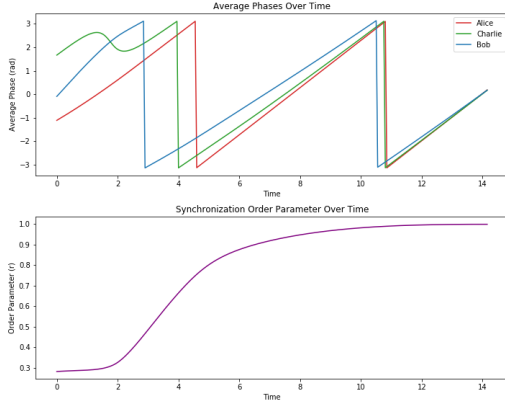


Fig. 8. Top: Average phase evolution for Alice (red), Charlie (green), and Bob (blue). Bottom: Order parameter $r(t)$ indicating global synchronization level.

D. GUI-Driven Interaction

A custom GUI implemented in Tkinter allows real-time tuning of inter-group coupling parameters:

- Sliders control K_{AC}^{forward} , K_{AC}^{backward} , K_{CB}^{forward} , and K_{CB}^{backward}
- Live rendering of network structure and phase-circle evolution

- Buttons for simulation start/stop and diagnostics plotting

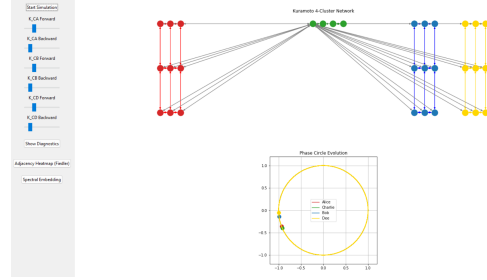


Fig. 9. Graphical user interface (GUI) for real-time simulation control and visual diagnostics. We can use this to visualise novel configurations such as the 4-cluster network seen here.

E. Insights and Applications

This simulation framework enables:

- Studying synchronization transitions in hierarchical networks
- Analyzing the effect of asymmetric and directional couplings
- Interactive exploration of small-world connectivity and delayed dynamics

F. Joint Spectral Embedding of the Four-Cluster Kuramoto Network

To analyse the structural relationships between oscillator clusters in the four-cluster Kuramoto network, we compute a two-dimensional *joint spectral embedding* using the second and third smallest eigenvectors of the combinatorial Laplacian of the undirected version of the graph. Let $L = D - A$ denote the Laplacian of the network, with D the diagonal degree matrix and A the adjacency matrix. Ordering the eigenvalues as

$$0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots,$$

the coordinates

$$(x_i, y_i) = (v_i^{(2)}, v_i^{(3)})$$

define a low-dimensional embedding of node i , where $v^{(k)}$ is the k th eigenvector of L . This embedding reveals how nodes group according to their *diffusion geometry*: nodes that have similar connectivity profiles obtain nearby coordinates in the spectral plane, while nodes belonging to different subnetworks tend to form clearly separated spectral clusters.

Figure 10 shows the two-dimensional spectral embedding for the Kuramoto network extended to 4 (Alice, Bob, Dee—each a 3×3 grid—and Charlie, representing

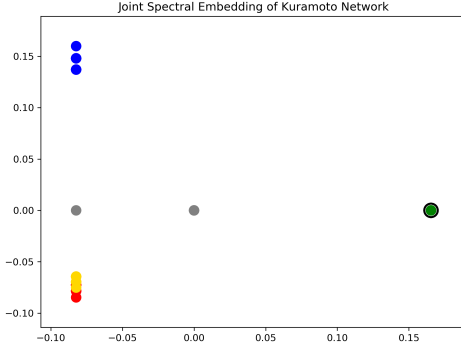


Fig. 10. **Joint spectral embedding of the four-cluster Kuramoto network.** Nodes are placed using the second and third eigenvectors of the graph Laplacian, revealing cluster geometry in diffusion space. Colours correspond to the four subnetworks: **Alice**, **Charlie**, **Bob**, and **Dee**. The **green Charlie nodes** occupy a central position, reflecting Charlie's role as the structural and dynamical hub that mediates all inter-cluster couplings.

the small-world core). Each point corresponds to one oscillator, coloured by cluster membership. The **green nodes** correspond specifically to the **Charlie** cluster. Their location in the interior of the spectral domain reflects the fact that Charlie serves as the structural hub of the full system: it is simultaneously close (in spectral geometry) to the red, blue, and yellow clusters, mirroring its role as the directional coupling center that connects all three peripheral grids.

G. Fiedler-Sorted Adjacency Matrix and Community Structure

To reveal large-scale community organization in the four-cluster Kuramoto network, we compute and visualise the *Fiedler-sorted adjacency matrix*. The method is based on the spectral properties of the graph Laplacian. Let L be the combinatorial Laplacian of the undirected version of the network. Denoting its eigenpairs by $(\lambda_k, v^{(k)})$, the *Fiedler vector* $v^{(2)}$ is the eigenvector associated with the second smallest eigenvalue [14], i.e.,

$$0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots$$

Ordering the nodes by the values of $v_i^{(2)}$ yields a one-dimensional reordering that places strongly interconnected groups of oscillators near one another.

Figure 11 shows the full weighted adjacency matrix of the Kuramoto network after sorting all nodes according to the Fiedler vector. Block-diagonal patterns become visible, corresponding to the four structural subnetworks: **Alice**, **Charlie**, **Bob**, and **Dee**. To assist visual interpretation, coloured membership bars on the top and left margins indicate the cluster identity of each node. The central Charlie cluster (green) appears tightly positioned

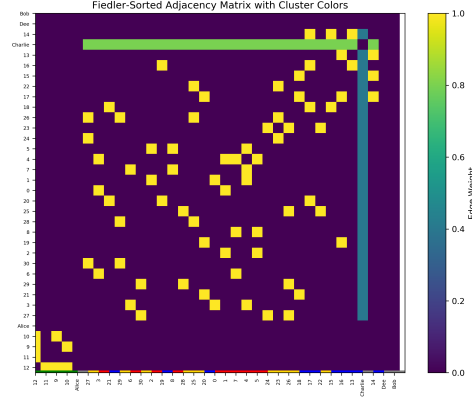


Fig. 11. **Fiedler-sorted adjacency matrix of the four-cluster Kuramoto network.** The adjacency matrix is reordered according to the Fiedler vector (second Laplacian eigenvector), revealing block structure that reflects the four subnetworks: **Alice**, **Charlie**, **Bob**, and **Dee**. Thin coloured bars along the top and left edges indicate node membership, allowing visual alignment of structural communities with the connectivity pattern. The strongly connected Charlie cluster occupies a spectrally central position, appearing between the peripheral clusters in the sorted ordering.

between the three peripheral clusters, reflecting its role as the dominant connector whose directional couplings mediate interactions among all other subnetworks.

XI. CONCLUSION

We have outlined a framework that unites power-law signaling, thermodynamic learning models, and quantum entanglement into a single metaheuristic interpretation grounded in the Kuramoto model. By extending the model to act as configuration spaces for entanglement condensates, we open a new way to simulate and understand multipartite quantum systems that can grow obeying network dynamics akin to classical counterparts, i.e. small-world, scale-free etc. This perspective also emphasizes synchronization as an emergent property of the networked quantum state, where entanglement acts as a phase coordination across topologically complex structures. Future directions include simulation of such networks using optical frameworks, experimental validation via photonic systems, and exploration of quantum network architectures built from this foundation. Indeed, one can conclude that a lot of the framework of adiabatic quantum processes could be at the very least simulated using such architecture of Kuramoto oscillators, perhaps emulating a lot of the synchronization and phase dynamics explored in harder to engineer and sustain entanglement systems e.g involving Josephson junctions or exotic optical states.

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DISCLOSURES

The authors declare no conflicts of interest.

DATA AVAILABILITY

Data and code related to this study are available from the corresponding author upon reasonable request.

Quantum-Kuramoto-Oscillator-Network-Graph-Simulation (Q-KONGS) source code available here: <https://github.com/MuonRay/Quantum-Kuramoto-Oscillator-Network-Graph-Simulation>

[This material is based upon work supported by Science Foundation Ireland (SFI) and is co-funded under the European Regional Development Fund under Grant Number 13/RC/2077 and 13/RC/2077-P2.]